

DengAI: predicting weekly dengue cases

Goal: estimate reported dengue cases each week in San Juan, and Iquitos using environmental measurements.

Load the data and take a first look

((1456, 24), (1456, 4), (416, 24))

1,456 historical weeks, 416 future weeks, and 24 input columns. The labels contain the city-week identifiers and the outcome.

Column	Row 1	Row 2
city	sj	sj
year	1990	1990
weekofyear	18	19
week_start_date	1990-04-30	1990-05-07
ndvi_ne	0.1226	0.1699
ndvi_nw	0.103725	0.142175
ndvi_se	0.198483	0.162357
ndvi_sw	0.177617	0.155486
precipitation_amt_mm	12.42	22.82
reanalysis_air_temp_k	297.573	298.211
reanalysis_avg_temp_k	297.743	298.443
reanalysis_dew_point_temp_k	292.414	293.951
reanalysis_max_air_temp_k	299.8	300.9
reanalysis_min_air_temp_k	295.9	296.4
reanalysis_precip_amt_kg_per_m2	32	17.94
reanalysis_relative_humidity_percent	73.3657	77.3686
reanalysis_sat_precip_amt_mm	12.42	22.82

Check the cities and their timelines

San Juan (sj) and Iquitos (iq) have different amounts of history.

city	count	min	max
iq	520	2000-07-01	2010-06-25
sj	936	1990-04-30	2008-04-22

city	count	min	max
iq	156	2010-07-02	2013-06-25
sj	260	2008-04-29	2013-04-23

Each city's test period follows its training period: approximately **five years** for San Juan and **three years** for Iquitos. We therefore validate on later weeks.

Next, we examine differences between cities.

The cities have different case levels and seasons

city	median	max
Iquitos	5	116
San Juan	19	461

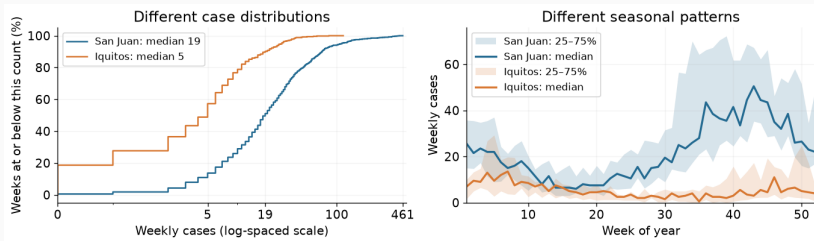


Figure 1: Case levels and seasonal timing differ between the two cities

The charts show differences in both the distribution of cases and the time of year when cases are higher.

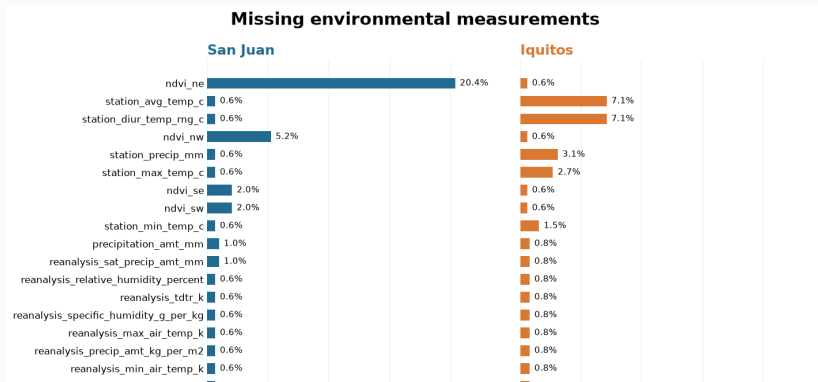
Decision: perform EDA separately, then prepare and fit a model for each city.

Missing Data

Missing measurements across all inputs:

ndvi_ne	194
ndvi_nw	52
station_diur_temp_rng_c	43
station_avg_temp_c	43
station_precip_mm	22
dtype: int64	

Vegetation (NDVI) have the most missing values



6. Weather history

Why look at earlier weather?

- **The effect may take time:** weather conditions today may be linked to dengue cases several weeks later. Earlier measurements help the model capture this delay.
- **Good conditions may build up:** our hypothesis is that several weeks of suitable warmth and moisture allow mosquito populations to grow more than one favourable week alone.
- **Duration matters:** rolling averages of temperature and humidity, and rainfall totals, describe whether conditions stayed favourable over several weeks.

What did we find and use?

- **In San Juan:** temperature and specific humidity have stronger positive associations with current cases around **8–10 observations earlier** than in the same week. The pattern differs in Iquitos.
- **Weather lags:** include measurements from **1, 2, 4, 8, 10, 20 and 26 observations earlier**.
- **Rolling summaries:** describe weather over the previous **4–26 observations** to capture sustained conditions.
- **Benchmark motivation:** DrivenData also suggests exploring earlier weather to address a mismatch in the timing of predicted cases. Benchmark discussion



7. Cross-validation

- **Keep time order:** train on earlier weeks and validate on later weeks.
- **Repeat five times per city:** predict the next **52 observations** (about one year), using more training history each round.
- **Keep a final check:** set aside the last **52 observations** as a separate holdout.
- **Choose settings using CV MAE:** lower average error is better. Then report the holdout error.
- **Why:** a random split would mix earlier and later periods. Chronological CV asks a question closer to the competition task.

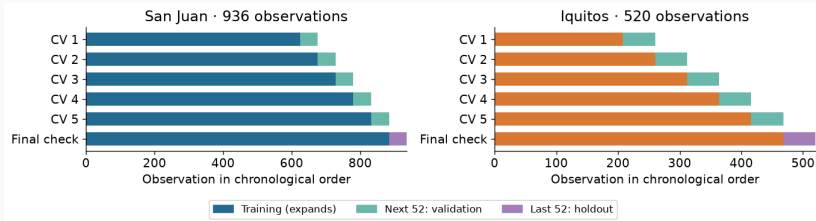


Figure 4: Chronological cross-validation timeline

8. Pipelines

- **One process:** fill gaps, add weather history, prepare inputs and train the model.
- **Avoid data leakage:** learn filling rules, scaling and feature selection from training data only.
- **Fair validation:** keep validation case counts out of training.
- **Consistent predictions:** reuse the same fitted preparation for new data.

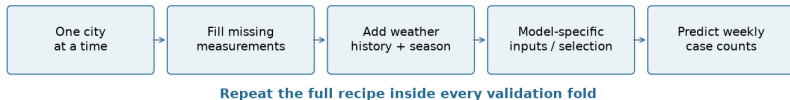


Figure 5: The modelling pipeline